A parity obstruction for Neumaier graphs with $e = s - 2$

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Abstract

A Neumaier graph is a non-complete edge-regular graph containing a regular clique; it is *strictly Neumaier* if it is not strongly regular. We study the class of Neumaier graphs whose parameters satisfy $e = s - 2$, where s is the size and e the nexus of the regular clique. We show that this condition forces the complement to decompose into $\binom{s}{2}$ classes of equal size indexed by the pairs of an s-set, so that the whole class is a two-parameter family indexed by (s, μ) with $\mu = v - 2k + \lambda$. Analysing the local structure we prove that the number of edges between two classes sharing an index is completely determined by the distribution of a certain perfect matching, and that the resulting integrality conditions give a parity obstruction: *if μ is odd then $4 \mid s$.*

In particular there is no Neumaier graph with parameters $(10\mu+5,\ 6\mu+4,\ 3\mu+3;\ 3, 5)$ for any odd μ. This answers a question of Abiad, De Boeck and Zeijlemaker, who observed computationally that these parameter sets are eliminated with unexpected ease and conjectured an underlying theoretical reason. Combining the obstruction with an exhaustive search for the remaining cases, we eliminate five further parameter sets left open in their table, namely $(28, 15, 6; 2, 4)$, $(34, 18, 7; 2, 4)$, $(40, 21, 8; 2, 4)$, $(45, 28, 15; 3, 5)$ and $(46, 24, 9; 2, 4)$. All evidence is consistent with the conjecture that the unique strictly Neumaier graph on 16 vertices is the only one with $e = s - 2$.

1 Introduction

Let Γ be a non-complete, non-empty graph. Γ is *edge-regular* with parameters (v, k, λ) if it has v vertices, is k-regular, and any two adjacent vertices have exactly λ common neighbours. A clique C of Γ is *e-regular* if every vertex outside C has exactly e neighbours in C, with $e > 0$. Following Neumaier [4], an edge-regular graph containing an e-regular clique of size s is called a *Neumaier graph* with parameters $(v, k, \lambda; e, s)$. Every strongly regular graph with a Delsarte clique is a Neumaier graph, and Neumaier asked in 1981 whether any Neumaier graph fails to be strongly regular. Such graphs are called *strictly Neumaier*; the first example, on 16 vertices with parameters $(16, 9, 4; 2, 4)$, was found by Evans, Goryainov and Panasenko [2], and is the unique smallest one.

Abiad, De Boeck and Zeijlemaker [1] determined the admissible parameter sets for strictly Neumaier graphs on at most 64 vertices and resolved several of them by a combination of structural arguments and integer programming. They close their paper with an observation they were unable to explain: their solver eliminated the parameter sets

$$(v, k, \lambda; e, s) \;=\; (10\mu + 5,\ 6\mu + 4,\ 3\mu + 3;\ 3, 5) \tag{1}$$

far faster than the size of the search space suggested, and they write that "this suggests that there may be an underlying theoretical reason why these graphs cannot exist for odd values of μ".

The purpose of this note is to supply that reason, in a form that applies far beyond (1). The family (1) is precisely the case $s = 5$ of the condition $e = s - 2$, and this condition turns out to be extremely rigid. Our main result is the following.

Theorem 1. *Let Γ be a Neumaier graph with parameters $(v, k, \lambda; e, s)$ satisfying $e = s - 2$, and set $\mu = v - 2k + \lambda$. If μ is odd then $4 \mid s$.*

Since $4 \nmid 5$, Theorem 1 eliminates the entire family (1) for odd μ in one stroke; the two instances $\mu = 3$ and $\mu = 5$ that required computation in [1] follow from a short parity count, as do all larger odd μ. The proof occupies Section 4 and uses no computer search.

Theorem 1 is silent when μ is even, and in Section 5 we report an exhaustive treatment of the smallest such parameter sets, which eliminates five parameter sets left blank in [1, Table 1]. Section 6 collects the remaining questions.

2 The reduction

Throughout, Γ is a Neumaier graph with parameters $(v, k, \lambda; e, s)$ and $e = s-2$, with e-regular clique $C = \{c_1, \dots, c_s\}$. We write

$$\Delta = \overline{\Gamma}, \qquad k' = v - k - 1, \qquad \mu = v - 2k + \lambda.$$

We record the standard translation to the complement.

Lemma 2. *Δ is k'-regular and μ-co-edge-regular, meaning that any two non-adjacent vertices of Δ have exactly μ common neighbours. The set C is a coclique of Δ in which every vertex outside C has exactly $s - e = 2$ neighbours. Moreover Γ is strictly Neumaier if and only if Δ is* not *edge-regular.*

Proof. Two vertices are non-adjacent in Δ exactly when they are adjacent in Γ, and for such a pair the number of common Δ-neighbours is $v-2-\big((k-1)+(k-1)-\lambda\big) = v-2k+\lambda = \mu$. A vertex outside C has e neighbours in C in Γ, hence $s - e = 2$ in Δ. Finally Γ is strongly regular precisely when it is co-edge-regular, i.e. when Δ is edge-regular. □

Because each vertex outside C has exactly two Δ-neighbours in C, the set $V(\Delta) \setminus C$ is partitioned by the pair of indices of those two neighbours. For a 2-subset $A = \{i, j\} \subseteq [s]$ put

$$P_A \;=\; N_\Delta(c_i) \cap N_\Delta(c_j).$$

As c_i and c_j are non-adjacent in Δ, they have exactly μ common neighbours, whence $|P_A| = \mu$ for every A. Counting vertices and the degree of c_i gives the following.

Proposition 3. *With the above notation,*

$$v = s + \binom{s}{2}\mu, \qquad k' = (s-1)\mu, \qquad k = v - 1 - (s-1)\mu, \qquad \lambda = \mu + 2k - v.$$

In particular the parameters of a Neumaier graph with $e = s - 2$ are determined by the pair (s, μ).

For example $(s, \mu) = (4, 2)$ gives $(16, 9, 4; 2, 4)$, the graph of [2], and $s = 5$ gives exactly the family (1). The parameter sets $(28, 15, 6; 2, 4)$, $(45, 28, 15; 3, 5)$, $(36, 25, 16; 4, 6)$, $(49, 36, 25; 5, 7)$ and $(64, 49, 36; 6, 8)$ of [1, Table 1] are the cases $(4, 4)$, $(5, 4)$, $(6, 2)$, $(7, 2)$ and $(8, 2)$.

3 Local structure

Fix $x \in P_A$ with $A = \{i, j\}$. Call a class P_B *near* x if $|A \cap B| = 1$ and *far* if $A \cap B = \varnothing$, and write $a_B(x) = |N_\Delta(x) \cap P_B|$. Set

$$T(x) \;=\; \sum_{B \cap A = \varnothing} a_B(x),$$

the number of far neighbours of x.

Lemma 4. *For every $k \notin A$,*

$$a_{\{i,k\}}(x) + a_{\{j,k\}}(x) \;=\; \mu - \varepsilon_k(x),$$

where $\varepsilon_k(x)$ is the number of far neighbours of x lying in a class whose index pair contains k.

Proof. Since $x \notin P_{\{i,k\}} \cup \dots$ we have $x \not\sim c_k$ in Δ, so $|N_\Delta(x) \cap N_\Delta(c_k)| = \mu$. Now $N_\Delta(c_k) = \bigcup_{l \neq k} P_{\{k,l\}}$. Of these classes, $P_{\{k,i\}}$ and $P_{\{k,j\}}$ are near x, and $P_{\{k,l\}}$ with $l \notin \{i, j, k\}$ is far from x; summing gives the claim. □

Lemma 5. *$a_A(x) \in \{\mu - 2, \mu - 1\}$ and $T(x) = a_A(x) - \mu + 2 \in \{0, 1\}$. Consequently the non-edges of Δ inside a class form a matching, and the far edges of Δ form a perfect matching on the set of vertices x with $T(x) = 1$.*

Proof. Summing Lemma 4 over the $s - 2$ indices $k \notin A$, each near class is counted once and each far class twice, so $\sum_{|B \cap A| = 1} a_B(x) + 2T(x) = (s - 2)\mu$. On the other hand x has two neighbours in C, so $a_A(x) + \sum_{|B \cap A| = 1} a_B(x) + T(x) = k' - 2 = (s - 1)\mu - 2$. Subtracting gives $a_A(x) - T(x) = \mu - 2$. Since $a_A(x) \leq \mu - 1$ we get $T(x) \leq 1$, and $T(x) \geq 0$ gives $a_A(x) \geq \mu - 2$. Thus x has at most one non-neighbour inside its own class, and exactly one far neighbour precisely when it has none. □

We call x *matched* if $T(x) = 0$ and *unmatched* otherwise, and write m_P for the number of matched vertices of the class P and $u_P = \mu - m_P$. By Lemma 5 the in-class non-edges pair up the matched vertices, so m_P is even and

$$u_P \equiv \mu \pmod 2 \qquad \text{for every class } P. \tag{2}$$

Lemma 6. *If $\{x, x'\}$ is a non-edge of Δ inside P_A, $A = \{i, j\}$, then*

$$N_\Delta(x) \cap N_\Delta(x') = \{c_i, c_j\} \cup (P_A \setminus \{x, x'\}).$$

In particular x and x' have no common neighbour outside P_A, and for each $k \notin A$ the sets $N_\Delta(x)$ and $N_\Delta(x')$ partition $P_{\{i,k\}} \cup P_{\{j,k\}}$.

Proof. The displayed set consists of common neighbours and has μ elements, and $x \not\sim x'$ forces exactly μ common neighbours; so there are no others. Both x and x' are matched, so $T(x) = T(x') = 0$ and Lemma 4 gives $|N_\Delta(x) \cap (P_{\{i,k\}} \cup P_{\{j,k\}})| = \mu$, and likewise for x'. Since the two sets are disjoint and their union has size 2μ, they partition. □

4 Forced edge counts and the parity obstruction

For disjoint classes P, Q let c_{PQ} be the number of edges of Δ between them; by Lemma 5 these are exactly the far edges, and they form a perfect matching on the unmatched vertices, so

$$\sum_{Q \cap P = \varnothing} c_{PQ} \;=\; u_P. \tag{3}$$

For $t \notin P$ put

$$f_P(t) \;=\; \sum_{\substack{Q \cap P = \varnothing \\ t \in Q}} c_{PQ},$$

the number of far edges leaving P towards a class whose index pair contains t. Finally let $E(A, B)$ denote the number of edges of Δ between the classes P_A and P_B.

Proposition 7. *Let $A = \{i, j\}$, $B = \{i, k\}$ and $D = \{j, k\}$ for distinct $i, j, k \in [s]$. Then*

$$E(A, B) \;=\; \tfrac{1}{2}\Big(\mu^2 - f_A(k) - f_B(j) + f_D(i)\Big). \tag{4}$$

Proof. Summing Lemma 4 over all $x \in P_A$, with the index k, gives $E(A, B) + E(A, D) = \mu^2 - f_A(k)$, since $\sum_{x \in P_A} \varepsilon_k(x)$ counts precisely the far edges from P_A to classes containing k. Applying the same identity to P_B with index j, and to P_D with index i, yields

$$E(A, B) + E(B, D) = \mu^2 - f_B(j), \qquad E(A, D) + E(B, D) = \mu^2 - f_D(i).$$

Adding the first two and subtracting the third gives $2E(A, B)$ as claimed. □

Since $E(A, B)$ is an integer, (4) forces the congruence

$$f_{\{i,j\}}(k) + f_{\{i,k\}}(j) + f_{\{j,k\}}(i) \;\equiv\; \mu^2 \;\equiv\; \mu \pmod 2 \tag{T_{ijk}}$$

for every 3-subset $\{i, j, k\} \subseteq [s]$; the left-hand side is symmetric in i, j, k, so there is one such congruence per triple. Together with (3) and (2),

$$\sum_{Q \cap P = \varnothing} c_{PQ} \;\equiv\; \mu \pmod 2 \qquad \text{for every class } P, \tag{D_P}$$

these are all we need.

Proof of Theorem 1. Suppose μ is odd, so that the right-hand side of every congruence (T_{ijk}) and (D_P) equals 1 modulo 2.

Step 1: s is even. Fix a pair $\{i, j\}$ and sum the congruences (T_{ijk}) over the $s - 2$ triples $\{i, j, k\}$ containing it. The three groups of terms behave as follows.

- $\displaystyle\sum_{k \notin \{i,j\}} f_{\{i,j\}}(k) = \sum_{Q \cap \{i,j\} = \varnothing} |Q|\, c_{\{i,j\}Q} = 2 \sum_{Q \cap \{i,j\} = \varnothing} c_{\{i,j\}Q} \equiv 0$, because every far class Q of $P_{\{i,j\}}$ contains exactly two indices, both outside $\{i, j\}$.

- $\displaystyle\sum_{k \notin \{i,j\}} f_{\{i,k\}}(j)$ is the sum of the $c_{\{i,a\},\{j,b\}}$ over all $a \notin \{i, j\}$ and $b \notin \{i, j, a\}$, since a far class of $P_{\{i,a\}}$ containing j has the form $\{j, b\}$ with $b \notin \{i, j, a\}$.

- $\displaystyle\sum_{k \notin \{i,j\}} f_{\{j,k\}}(i)$ is, by the same argument, the sum of the $c_{\{j,a\},\{i,b\}}$ over the same range of a, b. Interchanging the names of a and b identifies this with the previous sum, so the two cancel modulo 2.

Hence the whole left-hand side vanishes and we obtain $(s-2)\mu \equiv 0 \pmod 2$. As μ is odd, s is even.

Step 2: $4 \mid s$. Sum the congruences (D_P) over all $\binom{s}{2}$ classes. Every variable c_{PQ} occurs exactly twice, once for P and once for Q, so the left-hand side vanishes modulo 2 and $\binom{s}{2}\mu \equiv 0 \pmod 2$. As μ is odd this forces $\binom{s}{2} = \frac{s(s-1)}{2}$ to be even. By Step 1 the integer s is even, so $s-1$ is odd and therefore $\frac{s}{2}$ must be even, that is $4 \mid s$. □

Remark 8. The two steps are genuinely independent: for $s \equiv 3 \pmod 4$ the sum of *all* congruences (T_{ijk}) already gives $\binom{s}{3}\mu \equiv 0$ and suffices, and for $s \equiv 2 \pmod 4$ Step 2 alone suffices, but for $s \equiv 1 \pmod 4$ neither global sum is conclusive and the localised sum of Step 1 is needed. For $s = 5$ one may also verify the obstruction by the explicit 5-term certificate

$$D_{\{1,2\}} + D_{\{1,4\}} + D_{\{2,4\}} + D_{\{3,5\}} + T_{124} \;\equiv\; 0,$$

in which every variable cancels while the right-hand side is $5\mu \equiv \mu$.

Corollary 9. *There is no Neumaier graph with parameters* $(10\mu+5,\, 6\mu+4,\, 3\mu+3;\, 3, 5)$ *for any odd* μ*. Likewise there is none with parameters* $(15\mu+6,\, 10\mu+5,\, 6\mu+4;\, 4, 6)$ *or* $(21\mu+7,\, 15\mu+6,\, 10\mu+5;\, 5, 7)$ *for odd* μ*.*

Note that Theorem 1 constrains *all* Neumaier graphs with $e = s-2$, not merely the strict ones. This is consistent: when μ is odd and $s \not\equiv 0 \pmod 4$ the putative strongly regular complements independently fail standard feasibility conditions. For instance $(s, \mu) = (6, 1)$ would require a strongly regular graph with parameters $(21, 5, 1, 1)$, whose eigenvalue multiplicities are not integers.

5 The remaining cases

When μ is even, or when $4 \mid s$, Theorem 1 says nothing and the parameter sets must be treated individually. We did so by exhaustive search, encoding Δ in propositional logic. The search space is cut down by Lemmas 4–6 and by (4): one first enumerates the *skeletons*, that is the admissible data (m_P) and (c_{PQ}) up to the action of S_s, discarding those for which some right-hand side of (4) is not a non-negative integer; for each surviving skeleton the far matching may be normalised to the identity, the counts $E(A, B)$ are imposed as cardinality constraints, and the residual symmetry is broken lexicographically.

Strictness is imposed by a complete certificate. If Δ is edge-regular then the edges incident with C force the common-neighbour count to be $\lambda' = \mu s/2 - 2$, so Δ is edge-regular if and only if every edge has exactly λ' common neighbours; asserting that some edge violates this decides strictness outright, with no residual case analysis.

The results are collected in Table 1. The search was validated by recovering, in the case $(s, \mu) = (4, 2)$, exactly the unique strictly Neumaier graph of [2] together with exactly the two strongly regular graphs on the complementary parameters $(16, 6, 2, 2)$, namely the 4×4 rook's graph and the Shrikhande graph; and by reproducing the nonexistence results of [1] for $(4, 3)$, $(5, 2)$, $(5, 3)$. The case $(5, 4)$ required 988 skeletons, all of which were shown to admit no completion.

Combining Theorem 1 with Table 1 settles the family (1) throughout the range considered in [1].

Corollary 10. *There is no Neumaier graph with parameters* $(10\mu+5, 6\mu+4, 3\mu+3; 3, 5)$ *for any odd* μ*, nor for* $\mu \in \{2, 4\}$*. The smallest undecided member of the family is* $\mu = 6$*, on* 65 *vertices.*

(s,μ)	$(v,k,\lambda;e,s)$	Outcome	Source
$(4,2)$	$(16,9,4;2,4)$	unique strictly Neumaier graph	[2]
$(4,3)$	$(22,12,5;2,4)$	none	[1]
$(5,2)$	$(25,16,9;3,5)$	none	[1]
$(5,3)$	$(35,22,12;3,5)$	none	Theorem 1
$(5,5)$	$(55,34,18;3,5)$	none	Theorem 1
$(4,4)$	$(28,15,6;2,4)$	none	this note
$(4,5)$	$(34,18,7;2,4)$	none	this note
$(4,6)$	$(40,21,8;2,4)$	none	this note
$(4,7)$	$(46,24,9;2,4)$	none	this note
$(5,4)$	$(45,28,15;3,5)$	none	this note

Table 1: Status of the parameter sets with $e = s-2$ and $v \le 64$. The five entries marked "this note" are left blank in [1, Table 1].

6 Open problems

Conjecture 11. The graph of [2] on 16 vertices is the only strictly Neumaier graph with $e = s-2$.

Everything above is consistent with this, but the conjecture is open. We single out three concrete questions.

1. *A refinement of* (4) *modulo* 4. Theorem 1 uses only the parities of the forced counts $E(A,B)$, whereas (4) determines them exactly. A congruence modulo 4, or an inequality exploiting $0 \le E(A,B) \le \mu^2$ more aggressively, would be the natural route to the even-μ cases, which are currently settled only one parameter set at a time.

2. *The case* $4 \mid s$. Here the obstruction is vacuous for both parities of μ. The first open parameter set is $(64,49,36;6,8)$.

3. *A structural proof for* $s = 4$. By Lemma 6, matched pairs lying in classes that share an index are joined by a perfect matching, which endows each triangle of classes with a $\mathbb{Z}/2$ holonomy; a short computation shows that for each pair of classes sharing an index, exactly one triangle through a given pair of matched pairs has trivial holonomy. Making this rigidity yield a contradiction would replace the computations of Table 1 for $s = 4$ by an argument.

References

[1] A. Abiad, M. De Boeck and S. Zeijlemaker, *Neumaier graphs with few eigenvalues and the smallest strictly Neumaier graphs*, Graphs and Combinatorics **40** (2024).

[2] R. J. Evans, S. Goryainov and D. Panasenko, *The smallest strictly Neumaier graph and its generalisations*, Electronic Journal of Combinatorics **26** (2019), #P2.29.

[3] G. R. W. Greaves and J. H. Koolen, *Edge-regular graphs with regular cliques*, European Journal of Combinatorics **71** (2018), 194–201.

[4] A. Neumaier, *Regular cliques in graphs and special* $1\frac{1}{2}$*-designs*, in: Finite Geometries and Designs, London Mathematical Society Lecture Note Series **49**, Cambridge University Press, 1981, pp. 244–259.

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